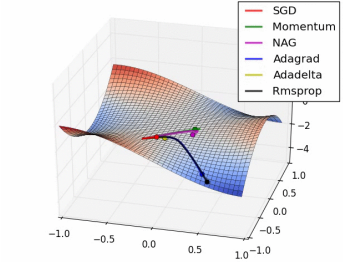
Optimization in Machine Learning

Advanced first order methods
Adam and friends

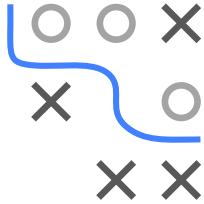


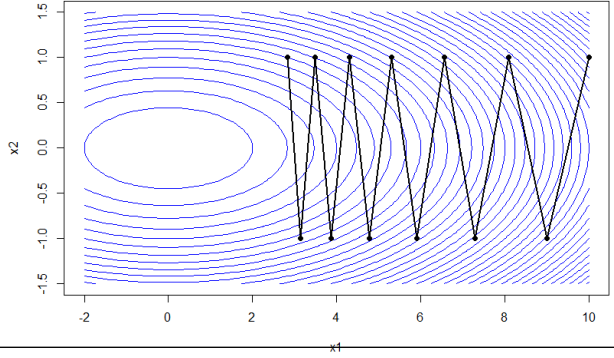
Learning goals

- Understand the idea of adaptive step size methods
- Understand the adaptive step size methods AdaGrad, RMSProp, and Adam and their differences

ADAPTIVE STEP SIZES

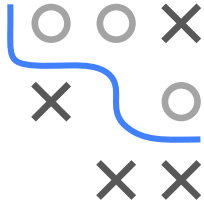
- Besides descent direction, step size is the other important control parameter in first order methods with strong influence on performance
- Motivated by limitations of global step sizes in poorly-conditioned problems, it is natural to use different step size for each dimension individually and automatically adapt sizes over time





ADAGRAD

- AdaGrad adapts step sizes by scaling them inversely proportional to square root of the sum of the past squared derivatives
  - Inputs with large derivatives get smaller step sizes
  - Inputs with small derivatives get larger step sizes
- This effectively changes the direction of the update (and not just the magnitude) by scaling the gradient element-wise

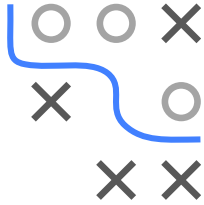


Algorithm AdaGrad

require Initial parameter $\boldsymbol{\theta}$, Global step size α, Small constant β for numerical stability (e.g. 10^{-7})
Initialize Gradient accumulation variable $\mathbf{r} = \mathbf{0}$
while stopping criterion not met **do**
    Sample minibatch of m examples from the training set $\{\tilde{\boldsymbol{x}}^{(1)}, \ldots, \tilde{\boldsymbol{x}}^{(m)}\}$
    Compute gradient estimate: $\hat{\mathbf{g}} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L\left(y^{(i)}, f\left(\tilde{\boldsymbol{x}}^{(i)} \mid \boldsymbol{\theta}\right)\right)$
    Accumulate squared gradient $\mathbf{r} \leftarrow \mathbf{r} + \hat{\mathbf{g}} \odot \hat{\mathbf{g}}$ (where $\odot$: element-wise product)
    Compute update: $\nabla\boldsymbol{\theta} = -\frac{\alpha}{\beta + \sqrt{\mathbf{r}}} \odot \hat{\mathbf{g}}$
    Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \nabla\boldsymbol{\theta}$
end while

RMSPROP

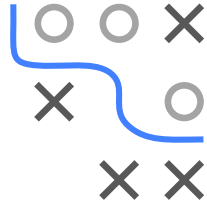
- AdaGrad's accumulation of squared gradients via summation can result in radically diminishing step sizes
- RMSProp is a modification of AdaGrad that resolves this by using an exponentially weighted moving average of past squared gradients
- Theoretically, RMSProp leads to performance gains in non-convex scenarios. Empirically, it is a very effective optimization algorithm routinely employed by DL practitioners.



Algorithm RMSProp

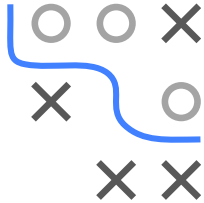
require Initial parameter $\boldsymbol{\theta}$, Global step size α, Small constant β for numerical stability (e.g. 10^{-7}), decay rate $\rho \in [0, 1)$
Initialize gradient accumulation variable $\mathbf{r} = \mathbf{0}$
while stopping criterion not met **do**
    Sample minibatch of m examples from the training set $\{\tilde{\boldsymbol{x}}^{(1)}, \ldots, \tilde{\boldsymbol{x}}^{(m)}\}$
    Compute gradient estimate: $\hat{\mathbf{g}} \leftarrow \frac{1}{m} \nabla_\theta \sum_i L\left(y^{(i)}, f\left(\tilde{\boldsymbol{x}}^{(i)} \mid \boldsymbol{\theta}\right)\right)$
    Accumulate squared gradient $\mathbf{r} \leftarrow \rho\,\mathbf{r} + (1-\rho)\,\hat{\mathbf{g}} \odot \hat{\mathbf{g}}$
    Compute update: $\nabla\boldsymbol{\theta} = -\frac{\alpha}{\beta + \sqrt{\mathbf{r}}} \odot \hat{\mathbf{g}}$
    Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \nabla\boldsymbol{\theta}$
end while

ADAM



- Adaptive Moment Estimation (ADAM) uses the 1st and 2nd moments of gradients
  - 1st moment: Exponentially decaying average of past gradients as momentum term
  - 2nd moment: Exponentially decaying average of past squared gradients to adapt step sizes like RMSProp
- Can thus be seen as combination of RMSProp and momentum

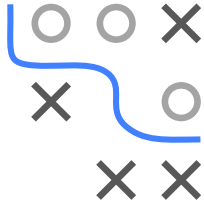
ADAM



Algorithm Adam

require Initial parameters $\boldsymbol{\theta}$, Global step size α (suggested default: 0.001), Small constant β (suggested default 10^{-8}), Exponential decay rates for moment estimates ρ_1 and ρ_2 in $[0, 1)$ (suggested defaults: 0.9 and 0.999 respectively)
Initialize time step $t = 0$
Initialize 1st and 2nd moment variables $\mathbf{s}^{[0]} = 0, \mathbf{r}^{[0]} = 0$
while stopping criterion not met **do**
    $t \leftarrow t + 1$
    Sample a minibatch of m examples from the training set $\{\tilde{x}^{(1)}, \ldots, \tilde{x}^{(m)}\}$
    Compute gradient estimate: $\hat{\mathbf{g}}^{[t]} \leftarrow \frac{1}{m} \nabla_{\boldsymbol{\theta}} \sum_i L\left(y^{(i)}, f\left(\tilde{\boldsymbol{x}}^{(i)} \mid \boldsymbol{\theta}\right)\right)$
    Update biased first moment estimate: $\mathbf{s}^{[t]} \leftarrow \rho_1 \mathbf{s}^{[t-1]} + (1 - \rho_1)\, \hat{\mathbf{g}}^{[t]}$
    Update biased second moment estimate: $\mathbf{r}^{[t]} \leftarrow \rho_2 \mathbf{r}^{[t-1]} + (1 - \rho_2)\, \hat{\mathbf{g}}^{[t]} \odot \hat{\mathbf{g}}^{[t]}$
    Correct bias in first moment: $\hat{\mathbf{s}} \leftarrow \frac{\mathbf{s}^{[t]}}{1-\rho_1^t}$
    Correct bias in second moment: $\hat{\mathbf{r}} \leftarrow \frac{\mathbf{r}^{[t]}}{1-\rho_2^t}$
    Compute update: $\nabla\boldsymbol{\theta} = -\frac{\alpha}{\beta + \sqrt{\hat{\mathbf{r}}}} \odot \hat{\mathbf{s}}$
    Apply update: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \nabla\boldsymbol{\theta}$
end while

ADAM: BIAS IN MOMENT ESTIMATES



- Initializes moment variables $\mathbf{s}$ and $\mathbf{r}$ with zero $\Rightarrow$ Bias towards zero

$$\mathbb{E}[\mathbf{s}^{[t]}] \neq \mathbb{E}[\hat{\mathbf{g}}^{[t]}] \quad \text{and} \quad \mathbb{E}[\mathbf{r}^{[t]}] \neq \mathbb{E}[\hat{\mathbf{g}}^{[t]} \odot \hat{\mathbf{g}}^{[t]}]$$

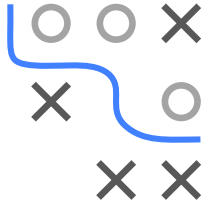
  ($\mathbb{E}$ calculated over minibatches)

- Indeed: Unrolling $\mathbf{s}^{[t]}$ yields

$$\begin{aligned}
\mathbf{s}^{[0]} &= 0 \\
\mathbf{s}^{[1]} &= \rho_1 \mathbf{s}^{[0]} + (1-\rho_1)\hat{\mathbf{g}}^{[1]} = (1-\rho_1)\hat{\mathbf{g}}^{[1]} \\
\mathbf{s}^{[2]} &= \rho_1 \mathbf{s}^{[1]} + (1-\rho_1)\hat{\mathbf{g}}^{[2]} = \rho_1(1-\rho_1)\hat{\mathbf{g}}^{[1]} + (1-\rho_1)\hat{\mathbf{g}}^{[2]} \\
\mathbf{s}^{[3]} &= \rho_1 \mathbf{s}^{[2]} + (1-\rho_1)\hat{\mathbf{g}}^{[3]} = \rho_1^2(1-\rho_1)\hat{\mathbf{g}}^{[1]} + \rho_1(1-\rho_1)\hat{\mathbf{g}}^{[2]} + (1-\rho_1)\hat{\mathbf{g}}^{[3]}
\end{aligned}$$

- Therefore: $\mathbf{s}^{[t]} = (1-\rho_1)\sum_{i=1}^{t} \rho_1^{t-i}\hat{\mathbf{g}}^{[i]}$
- **Note:** Contributions of past $\hat{\mathbf{g}}^{[i]}$ decrease rapidly

ADAM: BIAS CORRECTION



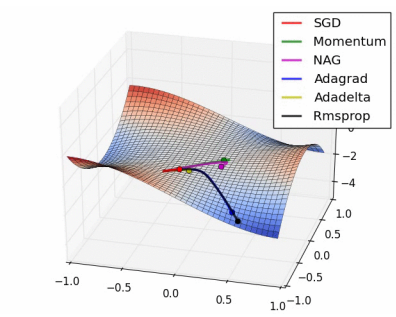
- We continue with

$$\begin{aligned}\mathbb{E}[\mathbf{s}^{[t]}] &= \mathbb{E}[(1-\rho_1)\sum_{i=1}^{t}\rho_1^{t-i}\hat{\mathbf{g}}^{[i]}] \\ &= \mathbb{E}[\hat{\mathbf{g}}^{[t]}](1-\rho_1)\sum_{i=1}^{t}\rho_1^{t-i}+\zeta \\ &= \mathbb{E}[\hat{\mathbf{g}}^{[t]}](1-\rho_1^t)+\zeta,\end{aligned}$$

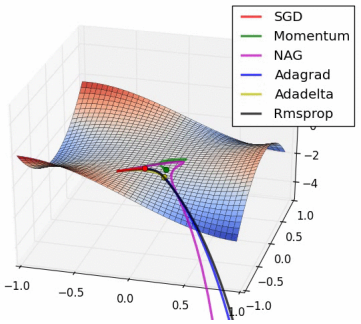
  where we approximated $\hat{\mathbf{g}}^{[i]}$ by $\hat{\mathbf{g}}^{[t]}$. The resulting error is put in ζ and is kept small due to the exponential weights of past gradients

- Therefore: $\mathbf{s}^{[t]}$ is a biased estimator of $\hat{\mathbf{g}}^{[t]}$
- But bias vanishes for $t \to \infty$ ($\rho_1^t \to 0$)
- Ignoring ζ, we correct for the bias by $\hat{\mathbf{s}}^{[t]} = \frac{\mathbf{s}^{[t]}}{(1-\rho_1^t)}$
- Analogously: $\hat{\mathbf{r}}^{[t]} = \frac{\mathbf{r}^{[t]}}{(1-\rho_2^t)}$

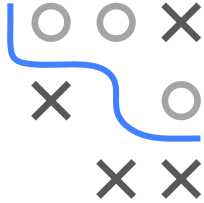
COMPARISON OF OPTIMIZERS: ANIMATION



▸ Click for source



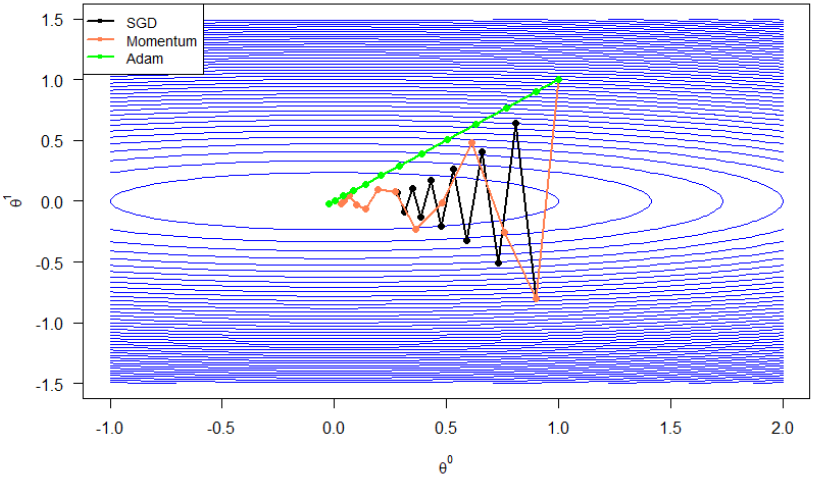
▸ Click for source



Comparison of SGD variants near saddle point
Left: After start. **Right:** Later
All methods accelerate compared to vanilla SGD
Best is RMSProp, then AdaGrad (Adam is missing here)

Credits: Dettmers (2015) and Radford

COMPARISON ON QUADRATIC FORM



SGD vs SGD with Momentum vs Adam on a quadratic form